

Homework 1 Pet Grooming Database Design

Management Information Systems Fall 2026

Course	33:136:370 Management Information Systems	Semester	Fall 2026
Instructor	Tripti Singh	Assignment	Homework 1
Value	3.75 points	Due	See Canvas

1 Assignment Overview and Purpose

This individual assignment asks you to convert repeating data from one flat spreadsheet into a clean relational database using Excel and MySQL Workbench.

Scope. Do not create a many-to-many relationship or a bridge table.

2 Learning Objectives

- Identify update, insertion, and deletion anomalies caused by repeated data.
- Use functional dependencies to decide where attributes belong.
- Decide the number of tables and choose appropriate table names, primary keys, foreign keys, datatypes, and one-to-many relationships.
- Create an EER model and an executable SQL script in MySQL Workbench.

3 Business Scenario and Rules

Paws Care Pet Grooming currently records all appointments in one Excel worksheet. Customer, pet, and service facts repeat, which makes updates unreliable. Your task is to redesign this information as a relational database.

- One customer may register many pets, but each pet belongs to one customer.
- One pet may have many appointments, but each appointment belongs to one pet.
- One service may appear in many appointments, but each appointment records exactly one service. If a pet receives another service, create another appointment row.

4 Source Data Flat Data Table

Create a source-data worksheet and enter all 10 records exactly as shown. Use one column for each field and one row for each appointment.

AppointmentID	AppointmentDate	CustomerID	CustomerName	CustomerPhone	PetID	PetName	ServiceID	ServiceName	StandardFee	Status
1001	2026-09-08	101	Maya Patel	201-555-0142	201	Coco	1	Bath and Brush	\$45.00	Completed
1002	2026-09-08	102	Liam Brooks	201-555-0188	202	Milo	3	Nail Trim	\$18.00	Completed
1003	2026-09-09	101	Maya Patel	201-555-0142	203	Luna	2	Full Groom	\$70.00	Completed
1004	2026-09-10	103	Sofia Rivera	201-555-0135	204	Max	1	Bath and Brush	\$45.00	Completed
1005	2026-09-11	104	Ethan Kim	201-555-0164	205	Bella	2	Full Groom	\$70.00	Scheduled
1006	2026-09-12	102	Liam Brooks	201-555-0188	202	Milo	1	Bath and Brush	\$45.00	Scheduled
1007	2026-09-13	103	Sofia Rivera	201-555-0135	206	Rocky	3	Nail Trim	\$18.00	Completed
1008	2026-09-14	101	Maya Patel	201-555-0142	201	Coco	2	Full Groom	\$70.00	Scheduled
1009	2026-09-15	104	Ethan Kim	201-555-0164	207	Nala	3	Nail Trim	\$18.00	Cancelled
1010	2026-09-16	103	Sofia Rivera	201-555-0135	204	Max	2	Full Groom	\$70.00	Scheduled

5 Tasks and Points

Task 1 Excel Workbook 0.50 points

Create LastName_FirstName_PetGrooming.xlsx. Include one worksheet containing the original source data and one additional worksheet for each normalized table in your design. You must decide how many normalized tables are needed and choose clear table and worksheet names. Do not lose any facts or relationships, and do not duplicate parent records.

Task 2 Analysis and Dependencies 0.65 points

Give one specific example each of an update anomaly, insertion anomaly, and deletion anomaly. List the functional dependencies in Determinant determines Dependent Attributes format. Draw a dependency bubble diagram with AppointmentID as the original row identifier.

Task 3 Relational Design and EER Model 0.95 points

Determine how many normalized tables are required. Choose each table name, primary key, foreign key or keys, attributes, MySQL datatypes, field lengths, required NOT NULL settings, and relationships. Build a complete EER diagram for your design.

Task 4 SQL Implementation 1.00 points

Create a database named paws_care. Write one executable SQL script that creates every table and constraint in your design in a valid order. Write INSERT statements that populate all tables using all facts from the 10 source rows. Add SELECT * statements for every table and run the complete script without errors.

Task 5 PDF Report 0.55 points

Create one PDF report containing the written analysis, dependency diagram, EER diagram, required screenshots, and captions listed on page 4.

Task 6 Submission Package 0.10 points

Upload exactly four separate files to Canvas: PDF, XLSX, SQL, and MWB. Use the required filenames and confirm that every submitted file opens.

6 Summary of Target Database Structure

Complete this planning table in your PDF report. You must decide how many tables are needed and enter each table name, primary key, foreign key or keys, attributes, and MySQL datatypes. Add or remove rows as needed.

Table Name	Primary Key	Foreign Keys	Attributes and Datatypes

Data requirement. After creating the design, populate every table using all facts represented by the 10 source records. No source record may be omitted.

7 Contents of the PDF Report

Order	Report Section	Required Content
1	Student information	Name, course, section, and Homework 1 title
2	Written analysis	Three anomaly examples and the functional dependency list
3	Flat Data evidence	Screenshot of Flat_Data with all headings and enough rows to show repetition
4	Dependency diagram	Readable bubbles or attributes and directional dependency arrows
5	EER evidence	Workbench EER screenshot showing all tables in your design, PK and FK markers, and relationships
6	SQL evidence	Screenshot showing successful execution and paws_care expanded with every created table
7	Data evidence	Screenshot or screenshots showing SELECT results for every table

8 Files Students Must Submit

Upload exactly four files separately to Canvas. Put all screenshots and diagrams inside the PDF report; do not upload separate image files and do not submit a ZIP file.

Required Filename	Required Contents
LastName_FirstName_HW1_Report.pdf	PDF report containing all seven sections above
LastName_FirstName_PetGrooming.xlsx	Original source worksheet plus one worksheet for each normalized table in the student design
LastName_FirstName_PetGrooming.sql	Executable CREATE, constraint, INSERT, and SELECT statements for paws_care
LastName_FirstName_PetGrooming.mwb	Native editable Workbench model containing the student-designed EER diagram

9 Grading Rubric

Criterion	Points	Full Credit Standard
Excel workbook	0.50	Source data is accurate; normalized worksheets contain complete, nonduplicated records and preserve all relationships.
Analysis and dependencies	0.65	Three specific anomaly examples, correct dependencies, and a readable dependency diagram.
Relational design and EER	0.95	The number of tables, table names, PKs, FKs, attributes, datatypes, constraints, and relationships form a sound design.
SQL implementation	1.00	paws_care and every designed table are created; all source facts are inserted; the script runs without errors.
PDF report	0.55	All required sections, diagrams, screenshots, and captions are readable and consistent with the native files.
Submission package	0.10	All four correctly named files are submitted separately and open successfully.
Total	3.75	Homework 1 represents 3.75 percent of the final course grade.